

AnyCam 3.0.0-rc1.5 Audit Report

Release audit - 02 October 2026 - scan Cancel, tests, the last module moves - tag 3.0.0-rc1.5

Scope confirmed by CrystalHeeler on 2026-10-02: (1) build plan B22, the scan's Cancel button. (2) Tests, then a move to its own file, for each part still in the main file: password entry, the snapshot loop, the Enhanced View engine, the manufacturer database, brand identification and the page builder (build plan E1). (3) The last type hints. Item 1 changes behaviour. Items 2 and 3 change no behaviour.

1. The scan's Cancel button (B22)

The Cancel endpoint set a flag named `_SCAN_CANCELLED` in `camera_discovery.py`. The scan read `SCAN_CANCELLED` in `anycam_scan.py`, a different name. Now there is one flag, `anycam_scan.SCAN_CANCELLED`, which the endpoint sets.

Where the scan checks it	What happens
Before the port scan (stage 2)	Stops; the status says 'Scan cancelled'.
Before each host, and before each port	No more probing.
Before each ONVIF-only device	No more probing.
At the end	'Scan cancelled - N device(s), M streaming.' The flag is cleared when the next scan starts.

5 checks (U14, U15): Cancel during the port probing stops it after the port in progress and probes no ONVIF-only device; Cancel during discovery runs no port scan; the next scan starts with the flag cleared.

2. The last type hints

30 functions had no type for an argument or the return value: the request argument of 19 web handlers, a log handler, an inner sort key and the background codec check in the main file, and 8 in `anycam_host.py`, `anycam_go2rtc.py`, `anycam_motion.py` and `anycam_scan.py`. Now every function in all 14 files is fully typed (an AST count: 0 missing). The code is unchanged apart from the annotations: each file's AST, with the annotations removed, is identical before and after.

3. Tests for the parts that moved

122 new checks in `tests/test_server.py`. Each section was written and passing before its code moved, and passing unchanged after. Stand-ins replace only what touches the network or a process: the ONVIF calls, the RTSP walkers, `ffprobe`, `ffmpeg` (a fake process that writes JPEGs) and the IEEE download. A local HTTP server plays a camera for the HTTP snapshot loop.

Section	Checks	What they pin
V	24	OUI key forms; lookup order (downloaded table, then the built-in camera and non-camera lists); camera or not; the cache read once; refresh only when 30 days old, its parse and save, a failed download; brand scoring (camera before NVR, word boundaries for short keywords); the RTSP realm rule; the generic entry; the brand throttle.
W	7	A complete page with every placeholder filled; the settings and the community address in the script (< escaped); <code>handle_index</code> builds the page once.

Section	Checks	What they pin
X	9	Unknown camera; classic entry clears a stale leave flag and the hardware failure count; the motion file reads the focus state (classic view: no second loop; live view: the loop restarts); classic leave stops the preheater and keeps the learned tier.
Y	42	ffmpeg arguments for card view, the settings and Enhanced View; JPEGs joined across reads; the idle stop; the transport flip after 3 empty runs and the codec clear after 5; the focus-leave flag; HTTP snapshots or ffmpeg; hardware decode and its software fallback; the quality ladder; the preheater; ffmpeg's error output (password removed, H.265+ flags); the H.265+ fallback; the snapshot endpoint in card and focus view; the status and log endpoints; HTTP Digest and query-string logins.
Z	40	Password entry by ONVIF (one-socket check at the RTSP port, ranking, duplicate dropped, identity kept, codec corrected in the background), by direct RTSP (port 554 first), for a rate-limited camera (stream table, locked streams, pacing), a wrong password, MJPEG, a DVR (one card per channel, the status endpoint); clear credentials; manual add; Deep Re-Probe (fresh, resumed, second stage, stale); the stream table.

Defects found while writing the tests

Both recorded in the build plan, not fixed: the scope had no behaviour change for these parts. No check pins either defect.

#	Defect	Effect
B23	api_add_camera makes the protocol upper case, then compares it with 'WebRTC'.	Adding a camera by hand with protocol WebRTC always answers 400 'Could not connect'. WS-RTSP works.
B24	_probe_db_streams paces a camera only if its STREAM_DB entry has throttle_type. No STREAM_DB entry has that key; it is in CAMERA_DB.	On a rate-limited camera (the Microseven) the stream-table check and each fprobe after it run without the 5 s wait. Reached only on the direct-RTSP path, or the ONVIF path with fewer than two working profiles.

4. The move

File	Content	Lines
anycam_credentials.py	Password entry, the stream table match and probe, the DVR channel list and its status endpoint, clear credentials, Deep Re-Probe, manual add	1,710
anycam_snap.py	snap_loop, http_snap_loop, the snapshot, status and log endpoints, the hardware preheater, ffmpeg's error output, the H.265+ fallback, the focus ladder	1,881
anycam_focus.py	The Enhanced View engine and the focus state (_FOCUSED_CAMERA, _FOCUS_ENGINE)	277
anycam_brand.py	The OUI database, the keyword tables and the loop that builds them, brand identification	471
anycam_page.py	build_html and handle_index	642

File	Content	Lines
camera_discovery.py	Settings, stores, the REST API, routing, start-up.	6,780 lines in 3.0.0-rc1.4 1,797

Link	Detail
The focus state	Only the focus functions replace it, so it moved with them. The main file, anycam_snap.py and anycam_motion.py read anycam_focus._FOCUSED_CAMERA. No file reads through H any more.
No import cycles	A split file imports another only if that file does not import it back, directly or through a third. Five names go through the main file's NEEDS for that reason: _MOTION (anycam_focus.py), _kill_hw_preheater (anycam_focus.py), _drain_stderr and _stop_proc (anycam_motion.py), _match_stream_db (anycam_probe.py).
Settings	The settings block (paths and options read from the environment) stays in the main file; the moved code takes those names through NEEDS.
NEEDS lists	117 names across all split files, checked by the gate.
The split tool (outside the repository) gained the rules above: no cycles, a moved value rewritten where other files read it, settings kept, a module-level loop moved with its tables, an import alias kept.	

Evidence that the move changed nothing

Check	Result
Every definition, text for text, after each of the 5 moves	363 before, 363 after; 0 missing, 0 changed, apart from the anycam_focus. prefix on the focus state. 124 in the main file.
Built page and routes, after each move	Same page hash; the same 56 routes.
Tests, after each move	448 of 448 server checks, 114 of 114 page checks.
Main file tidy-up	265 lines of empty banners, stale comments and blank runs removed; the file's AST identical before and after.
Packaged zip	Imported from its own folder.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 14 files compiled; module list matches the imports and the Dockerfile; 212 top-level definitions; 48 contracts; 117 NEEDS names; build inputs checked online.
Python integration	PASS	448 of 448, 122 new (and 5 for B22).
JavaScript behaviour	PASS	114 of 114.
Undefined names	PASS	14 files, 0.
Not run	NOTED	The image was not built and nothing ran on a camera from the build PC. The tests use stand-ins for ffmpeg, fprobe and the probers. The field test (A19) is the first run of the moved code.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.0.0-rc1.4 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	14 files.
1.2 Syntax	compile() per file	PASS	14 files.
1.3 Runtime errors	Names	PASS	B22 fixed: one cancel flag.
1.6 Function design	File size	NOTED	Main file 1,797 lines. snap_loop (1,048 lines) and api_set_credentials (865) moved whole; splitting them is a behaviour risk and was not in scope.
1.7 Mutability	Values replaced at run time	PASS	The focus state has one owner file; the gate enforces the import rule.
1.14 Type checking	AST count	PASS	0 functions missing a type.
1.16 Refactoring	Behaviour kept	PASS	Text comparison of 363 definitions after each move; page hash; routes; tests before and after.
1.18 Testing	New tests	PASS	122 checks, written before the moves; they found B23 and B24.
6.3 Version control	Commits and gate	PASS	One commit for each part; the gate run before the release commit.
Rule 9 (privacy)	New text	PASS	Tests and documents use made-up addresses only.

Known issues carried forward

- **B23**: adding a camera by hand with protocol WebRTC fails.
- **B24**: the stream-table check during password entry does not pace rate-limited cameras.
- **B20** (late pictures, insects on the fallback): right after 3.0.0.
- **C17** (detection zones): design discussion pending.
- Skip Non-Reference Frames breaks H.264 cameras (B3).

Acceptance (field test A19)

- The add-on starts; the start-up scan finds the same cameras as before.
- Cancel during a scan stops it, and the status says 'Scan cancelled'.
- A camera password is accepted; on test system B the DVR channel cards appear.
- Cards show pictures; Enhanced View opens in live and in classic view.

- Motion detection still records.
- The log has no line with 'not defined' or 'has not run yet'.

Files changed

File	Change
camera_discovery.py	Five parts moved out; the cancel flag; type hints; empty banners removed. Version 3.0.0-rc1.5.
anycam_credentials.py, anycam_snap.py, anycam_focus.py, anycam_brand.py, anycam_page.py	New.
anycam_scan.py	The cancel flag and its checks; imports from anycam_brand.
anycam_motion.py	Reads anycam_focus._FOCUSED_CAMERA; type hints.
anycam_probe.py	Imports from anycam_brand.
anycam_host.py, anycam_go2rtc.py	Type hints; docstring.
anycam_modules.py, Dockerfile	Five more files.
tests/	127 server checks; run_tests.py builds the page from anycam_page.py.
docs, CLAUDE.md, README	Module list, no-cycle rule; build plan.
CHANGELOG.md, config.yaml	3.0.0-rc1.5.